Vertiport Planning in Chicago

Mathematical model, working document v1.1

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Purpose and attribution

This document states the full optimization model used in the project. The modeling framework (facility location with endogenous passenger demand, and the distributionally robust demand function) is due to **Wang, Jacquillat and Vaze (2022)**, "Vertiport Planning for Urban Aerial Mobility: An Adaptive Discretization Approach," *M&SOM* 24(6):3215–3235. This project implements a simplified version (no operations layer) and adds two questions:

- **RQ1.** What happens when customer expectations are estimated per origin–destination pair from *observed* rideshare fares and travel times, instead of being set uniformly?
- **RQ2.** How robust is the vertiport network to time-of-day demand patterns, and what does a network planned for a single period lose?

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1 Overview

The model has two layers. A *precomputation* layer (Python) turns trip data into travel times and into the share of travelers who would switch to an air taxi for each possible routing. All nonlinear relations (the switching-share functions and the estimation of customer expectations) live there. An *optimization* layer (a mixed-integer linear program solved with Gurobi) then decides where to build vertiports and how to route each origin–destination flow. The reported results maximize the number of air-taxi passengers; the profit version is the same model with the per-passenger margins of Section 4.5 as weights. Figure 1 shows the pipeline.

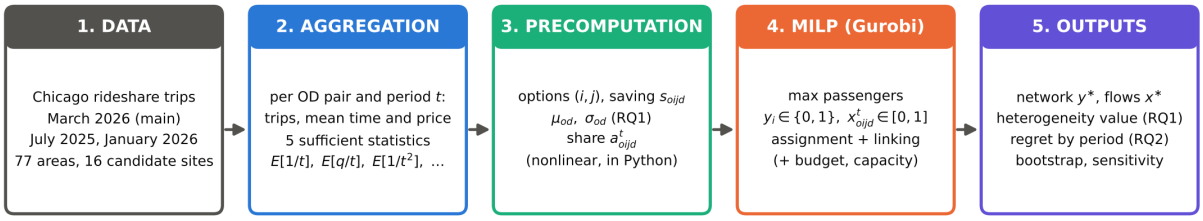


Figure 1: Pipeline from raw trip records to the optimal network and the RQ1/RQ2 metrics (t: ride time, q: ride price, s: relative time saving, a: share of travelers who switch).

Figure 2 shows the decision faced for a single origin–destination (OD) pair in a single period: each traveler either stays on the ground (rideshare) or flies through an open pair of vertiports (i, j).

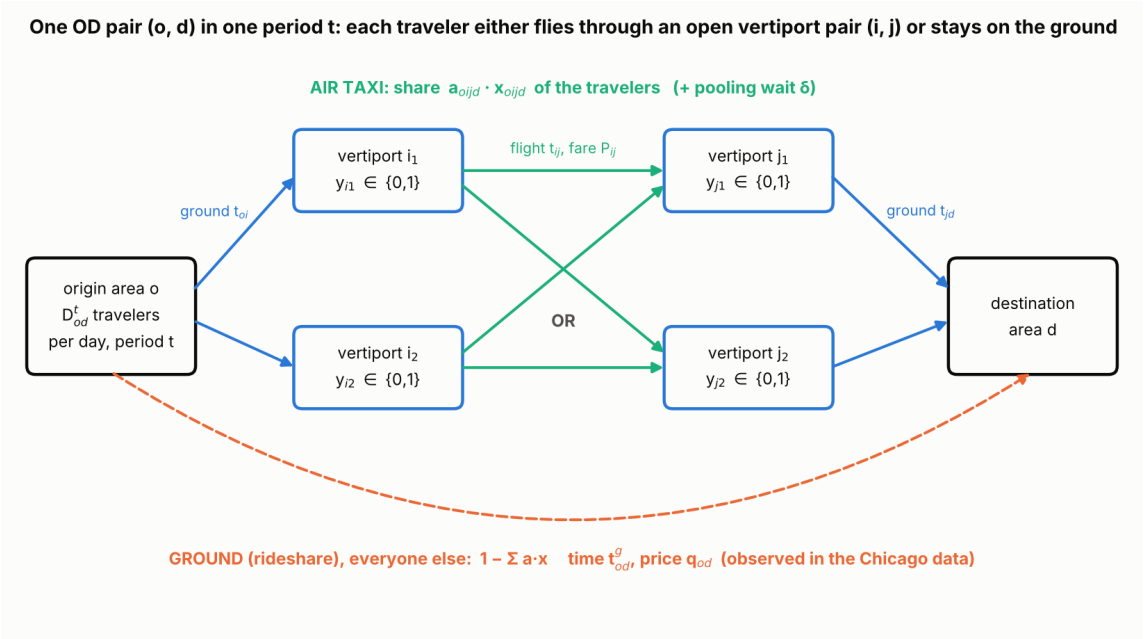


Figure 2: Routing options for one OD pair (o, d) in period t. Blue: ground legs to and from vertiports; green: flight legs; orange: travelers who stay on the ground.

2 Sets and indices

Symbol	Index	Meaning
$\mathcal{Z}$	o, d	Zones: the 77 Chicago community areas (origins and destinations).
$\mathcal{V} \subseteq \mathcal{Z}$	i, j	Candidate vertiport sites (e.g. O'Hare, Midway, Vertiport Chicago, Loop, Near North Side).
$\mathcal{T}$	t	Periods: $\{\text{am}, \text{mid}, \text{pm}, \text{we}, \text{oth}\}$ (7–10, 10–16, 16–19 on weekdays; weekend; other hours).
$\mathcal{P} \subseteq \mathcal{Z} \times \mathcal{Z}$	(o, d)	OD pairs kept: $o \neq d$, ground time $\geq t_{\min}$, at least $n_{\min}$ trips per day.
$\mathcal{K}_{od} \subseteq \mathcal{V} \times \mathcal{V}$	(i, j)	Useful vertiport pairs for OD pair (o, d) (Section 4.2).

3 Data and parameters

3.1 Observed in the Chicago rideshare data

Source: City of Chicago, Transportation Network Providers trips, March 2026 (Socrata `6dvr-xwnh`). Shared trips are removed; ground time uses `trip_seconds` (start and end timestamps are rounded to 15 minutes); ground price is `fare` + `additional_charges` (fares are rounded to $2.50).

Symbol	Meaning	Unit
D^t_{od}	Rideshare trips from o to d in period t, per day of that period type	trips/day
$t^{g,t}_{od}$	Mean ground travel time from o to d in period t	min
q_k, t_k	Price and ground time of an individual trip k	\$, min
$m^{(1..5),t}_{od}$	Sufficient statistics $\mathbb{E}[1/t]$, $\mathbb{E}[q/t]$, $\mathbb{E}[1/t^2]$, $\mathbb{E}[q/t^2]$, $\mathbb{E}[q^2/t^2]$ (Section 5)	various
ℓ_{od}, ℓ_{ij}	Straight-line distance between zone centroids	miles

3.2 Assumed (with sensitivity ranges)

The online appendix of Wang et al. (2022) was not available; assumed values come from public sources (see the data-sources note in the project).

Symbol	Meaning	Default	Range
v	eVTOL block speed	200 km/h	150–240
t_{oh}	Fixed overhead per flight (boarding, takeoff, landing)	10 min	6–20
$\bar{\delta}$	Pooling wait (fixed; endogenous in the paper)	5 min	2–10
t_0	Ground leg when trip starts or ends in the vertiport's zone	5 min	3–8
h	Fixed transfer ("hassle") penalty added to the air-taxi time (v1.1)	0 min	0–15
$s_{\min}$	Minimum relative time saving for an option to be offered (v1.1)	0.15	0–0.15
$t_{\max}$	Maximum ground leg to or from a vertiport	25 min	15–30
k	Vertiports considered at each end of a trip	3	2–5
μ	Mean required relative time saving (uniform case)	0.40	0.25–0.60
σ	Spread of required relative time saving (uniform case)	0.33	0.20–0.40
b	Value of travel time	\$21.80/h	\$19–65/h
P_0, r	Air-taxi fare: base and rate per passenger-mile	\$15, \$4.50	\$10–20, \$2–9
κ	Flight operating cost per passenger-mile	to be set	sensitivity
g_i	Daily cost of a vertiport at site i	to be set	sensitivity
w_t	Number of days per year of period type t	250 (weekday), 115 (weekend)	–

4 Precomputation

4.1 Ground and flight times

Ground legs. For a ground leg between zones u and v in period t:

$$t^t_{uv} = \begin{cases} t_0 & \text{if } u = v, \\ \text{observed mean } t^{g,t}_{uv} & \text{if at least } n_{\min} \text{ trips per day}, \\ \alpha^t + \beta^t \, \ell_{uv} & \text{otherwise (linear fit on well-observed pairs).} \end{cases} \tag{1}$$

Flight legs. With the haversine distance ℓ_{ij} (km for this formula):

$$t^f_{ij} = \frac{60 \, \ell_{ij}}{v} + t_{\text{oh}}. \tag{2}$$

4.2 Useful vertiport pairs

Let $N^{\text{acc},t}(o)$ be the k candidate sites with the shortest ground leg $t^t_{oi} \le t_{\max}$ from o, and $N^{\text{egr},t}(d)$ the k sites with the shortest leg $t^t_{jd} \le t_{\max}$ to d. Then

$$\mathcal{K}^t_{od} = \left\{ (i,j) \in N^{\text{acc},t}(o) \times N^{\text{egr},t}(d) \;:\; i \neq j \right\}. \tag{3}$$

4.3 Air-taxi trip time and time saving

As in equation (5) of the paper, with the endogenous pooling and detour times replaced by a fixed pooling wait:

$$\tau^t_{oijd} = t^t_{oi} + \bar{\delta} + t^f_{ij} + t^t_{jd} + h, \tag{4}$$

$$s^t_{oijd} = 1 - \frac{\tau^t_{oijd}}{t^{g,t}_{od}} \qquad \text{(relative time saving)}. \tag{5}$$

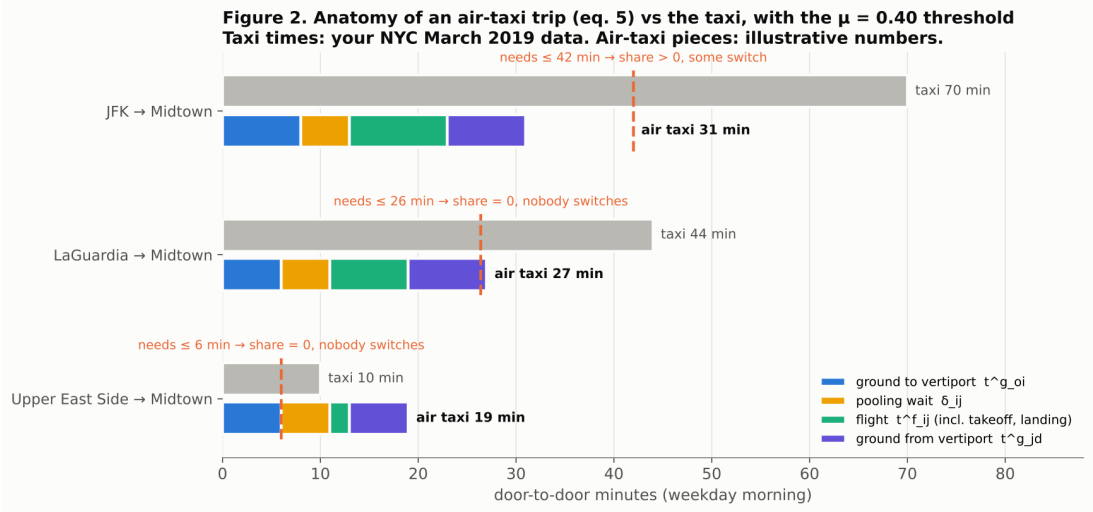


Figure 3: Anatomy of an air-taxi trip compared with the ground trip (NYC example from the validation step). Only long trips leave enough margin after the fixed components.

4.4 Demand: the share of travelers who switch

Following Section 4.4 of the paper, a traveler on OD pair (o, d) compares utilities $V - P_{od} - b\,\tau$ (air taxi) and $V - q_{od} - b\,t^g_{od}$ (ground). She flies if and only if

$$\tau \le (1-\theta)\,t^g_{od} \quad \Longleftrightarrow \quad \theta \le s, \qquad \text{where} \quad \theta = \frac{P_{od} - q_{od}}{b\,t^g_{od}} \tag{6}$$

is the *required relative time saving*: the price premium of flying, converted into minutes and divided by the ground time. In all formulas with times in minutes, b is expressed in dollars per minute ($b = 21.80/60$ in the base case). Across travelers, θ is random with mean μ and standard deviation σ. The share who fly is $\Pr(\theta \le s)$.

Distributionally robust share (Proposition 1 of the paper). When only μ and σ are known, the worst case over all distributions with these two moments is given by the one-sided Chebyshev (Cantelli) inequality:

$$\inf_{F:\ \mathbb{E}_F\theta=\mu,\ \mathrm{Var}_F\theta=\sigma^2} \Pr_F(\theta \le s) = \begin{cases} \dfrac{(s-\mu)^2}{\sigma^2 + (s-\mu)^2} & s \ge \mu, \\ 0 & s < \mu. \end{cases} \tag{7}$$

Writing $\tilde\tau = (s-\mu)/\sigma$, the share used in the model is

$$\boxed{a^t_{oijd} = \frac{\tilde\tau^2}{1+\tilde\tau^2} \ \text{ if } \tilde\tau > 0, \qquad a^t_{oijd} = 0 \text{ otherwise}, \qquad \tilde\tau = \frac{s^t_{oijd} - \mu_{od}}{\sigma_{od}}.} \tag{8}$$

In the uniform case $\mu_{od} = \mu$ and $\sigma_{od} = \sigma$ for all OD pairs. In RQ1 they are estimated from data (Section 5). Because a is computed before optimization, the optimization model stays linear. In the paper, by contrast, τ depends on decisions (pooling and detours), so the S-curve sits inside the model and requires adaptive discretization.

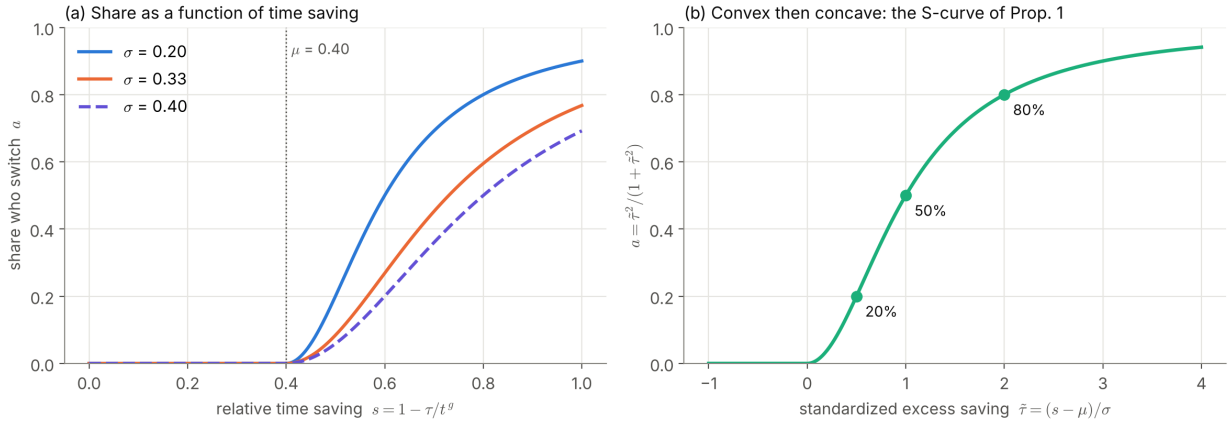


Figure 4: The demand function (8). (a) Nobody switches until the saving exceeds μ; a larger σ flattens the curve. (b) In standardized units the curve is convex then concave: 20%, 50% and 80% of travelers switch at $\tilde\tau = 0.5, 1, 2$.

4.5 Fares and margins

The air-taxi fare is quoted door to door on the OD distance; the operator pays the flight and the two ground legs:

$$P_{od} = P_0 + r\,\ell_{od}, \tag{9}$$

$$c_{oijd} = \kappa\,\ell_{ij} + c^g\,(t^t_{oi} + t^t_{jd}) \qquad \text{(flight cost + ground-leg cost)}, \tag{10}$$

$$\pi_{oijd} = P_{od} - c_{oijd} \qquad \text{(margin per air-taxi passenger)}. \tag{11}$$

Setting $\pi_{oijd} = 1$ for all options turns profit into the number of passengers carried, which is the objective used for the NYC validation.

5 RQ1: estimating customer expectations from trip data

For each trip k observed on OD pair (o, d) in period t, with price q_k and ground time t_k:

$$\theta_k = \frac{P_{od} - q_k}{b\, t_k}. \tag{12}$$

Sufficient statistics. Downloading every trip for every point of the fare and value-of-time grid would be slow. Instead, Chicago's server returns five averages per (o, d, t):

$$m_1 = \mathbb{E}\left[\tfrac{1}{t}\right], \quad m_2 = \mathbb{E}\left[\tfrac{q}{t}\right], \quad m_3 = \mathbb{E}\left[\tfrac{1}{t^2}\right], \quad m_4 = \mathbb{E}\left[\tfrac{q}{t^2}\right], \quad m_5 = \mathbb{E}\left[\tfrac{q^2}{t^2}\right]. \tag{13}$$

Since θ is linear in P and q once divided by $b\,t$, its first two moments follow exactly for *any* fare P_{od} and value of time b:

$$\mu_{od} = \mathbb{E}[\theta] = \frac{P_{od}\, m_1 - m_2}{b}, \tag{14}$$

$$\mathbb{E}[\theta^2] = \frac{P_{od}^2\, m_3 - 2P_{od}\, m_4 + m_5}{b^2}, \tag{15}$$

$$\sigma_{od}^2 = \mathbb{E}[\theta^2] - \mu_{od}^2. \tag{16}$$

Fare rounding correction. Fares are rounded to the nearest \$2.50. If the rounding error ε is uniform on $[-1.25, 1.25]$ and independent of t, it adds $\mathrm{Var}(\varepsilon)\,\mathbb{E}[1/t^2]/b^2$ to $\mathrm{Var}(\theta)$, so

$$\sigma_{od}^{2,\mathrm{corr}} = \sigma_{od}^2 - \frac{2.5^2}{12} \cdot \frac{m_3}{b^2}. \tag{17}$$

Thin OD pairs (optional shrinkage). With n_{od} trips, shrink toward the demand-weighted mean $\bar{\mu}$: $\hat{\mu}_{od} = \frac{n_{od}\,\mu_{od} + \lambda\,\bar{\mu}}{n_{od} + \lambda}$ (same for σ), with λ chosen by cross-validation or set to 20.

Fair uniform benchmark. The uniform case uses the demand-weighted averages

$$\bar{\mu} = \frac{\sum_{od} D_{od}\,\mu_{od}}{\sum_{od} D_{od}}, \qquad \bar{\sigma} = \frac{\sum_{od} D_{od}\,\sigma_{od}}{\sum_{od} D_{od}}, \tag{18}$$

so the two cases differ only by heterogeneity, not by the average level.

The grid. Because realistic fares give $\theta > 1$ on many OD pairs (nobody switches), results are reported over a grid $\mathcal{G}$ of fare rates r and values of time b (Figure 5).

6 Decision variables

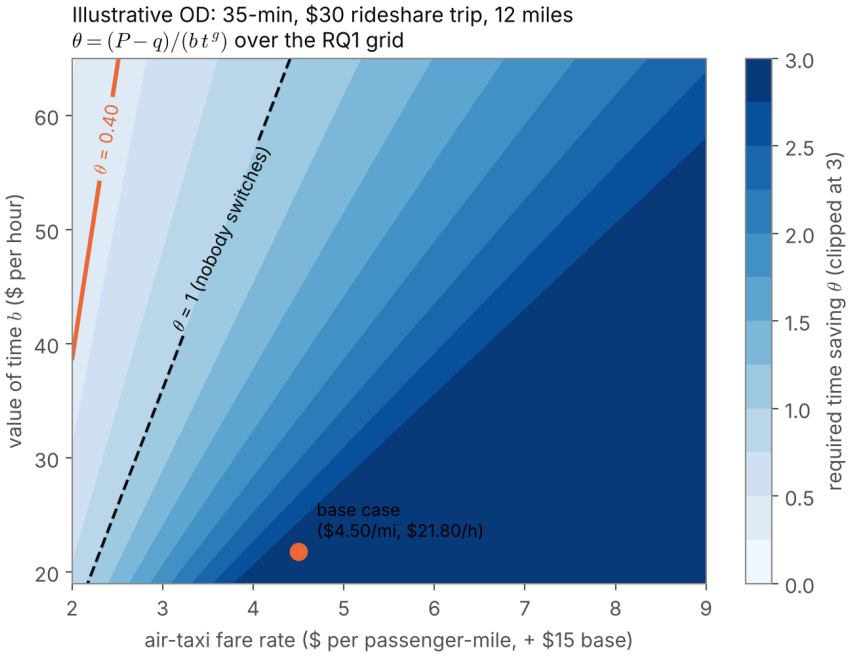


Figure 5: Required time saving θ for an illustrative OD pair over the RQ1 grid. At the base case ($4.50/mile, $21.80/h), $\theta > 2$: no traveler on this pair switches. Only cheap fares or high values of time (airport travelers) bring θ near 0.40.

Decision variables

$$y_i = \begin{cases} 1 & \text{if a vertiport is built in zone } i \\ 0 & \text{otherwise} \end{cases} \qquad i \in \mathcal{V}$$

x^t_{oijd} = share of the travelers from o to d in period t who are assigned to the vertiport pair $i \to j$ $\qquad (o,d) \in \mathcal{P},\ (i,j) \in \mathcal{K}^t_{od},\ t \in \mathcal{T}$

The number of air-taxi passengers on option (o,i,j,d) in period t is $D^t_{od}\, a^t_{oijd}\, x^t_{oijd}$ per day.

7 Optimization models

7.1 Model P0: replication of Table 4 (NYC validation)

Model P0: passengers minus a cost per vertiport

$$\max \quad \sum_{(o,d)\in\mathcal{P}} \sum_{(i,j)\in\mathcal{K}_{od}} D_{od}\, a_{oijd}\, x_{oijd} \;-\; c \sum_{i\in\mathcal{V}} y_i \tag{19}$$

$$\text{s.t.} \quad \sum_{(i,j)\in\mathcal{K}_{od}} x_{oijd} \le 1 \qquad \forall (o,d)\in\mathcal{P} \tag{20}$$

$$\sum_{j:(i,j)\in\mathcal{K}_{od}} x_{oijd} \le y_i \qquad \forall (o,d)\in\mathcal{P},\ i\in\mathcal{V} \tag{21}$$

$$\sum_{i:(i,j)\in\mathcal{K}_{od}} x_{oijd} \le y_j \qquad \forall (o,d)\in\mathcal{P},\ j\in\mathcal{V} \tag{22}$$

$$y_i \in \{0,1\}, \quad 0 \le x_{oijd} \le 1 \qquad \forall\, i,\ (o,i,j,d) \tag{23}$$

Here c is a cost per vertiport in passenger units, calibrated so that $\mu = 0.40$ gives the paper's three vertiports; μ is then swept from 0.40 to 0.33 with c fixed (results in Appendix A).

7.2 Model P1: single-period profit model (Chicago base case)

Model P1: maximize daily operating profit for one period

$$\max \quad \sum_{(o,d)\in\mathcal{P}} \sum_{(i,j)\in\mathcal{K}_{od}} \pi_{oijd}\, D_{od}\, a_{oijd}\, x_{oijd} \;-\; \sum_{i\in\mathcal{V}} g_i\, y_i \tag{24}$$

$$\text{s.t.} \quad \sum_{(i,j)\in\mathcal{K}_{od}} x_{oijd} \le 1 \qquad \forall (o,d)\in\mathcal{P} \tag{25}$$

$$\sum_{j:(i,j)\in\mathcal{K}_{od}} x_{oijd} \le y_i \qquad \forall (o,d)\in\mathcal{P},\ i\in\mathcal{V} \tag{26}$$

$$\sum_{i:(i,j)\in\mathcal{K}_{od}} x_{oijd} \le y_j \qquad \forall (o,d)\in\mathcal{P},\ j\in\mathcal{V} \tag{27}$$

$$\sum_{i\in\mathcal{V}} y_i \le p \tag{28}$$

$$\frac{1}{H} \sum_{(o,d)\in\mathcal{P}} \sum_{(i',j')\in\mathcal{K}_{od}:\ i\in\{i',j'\}} D_{od}\, a_{oi'j'd}\, x_{oi'j'd} \;\le\; Q\, y_i \qquad \forall i\in\mathcal{V} \tag{29}$$

$$y_i \in \{0,1\}, \quad 0 \le x_{oijd} \le 1 \qquad \forall\, i,\ (o,i,j,d) \tag{30}$$

Constraint	Meaning
(25) Assignment	each OD pair is routed through at most one combination in total; the rest stays on the ground
(26) Departure link	travelers can only depart from a vertiport that is built
(27) Arrival link	travelers can only arrive at a vertiport that is built
(28) Budget	at most p vertiports (optional; used for the passengers-versus-p curves)
(29) Capacity	departing plus arriving passengers per hour at site i are at most Q (H = hours in the period; optional)

7.3 Model P2: multi-period robust network (RQ2)

Model P2: one network for all periods

$$\max \quad \sum_{t\in\mathcal{T}} w_t \sum_{(o,d)\in\mathcal{P}} \sum_{(i,j)\in\mathcal{K}^t_{od}} \pi^t_{oijd}\, D^t_{od}\, a^t_{oijd}\, x^t_{oijd} \;-\; G\sum_{i\in\mathcal{V}} g_i\, y_i \tag{31}$$

$$\text{s.t.} \quad \sum_{(i,j)\in\mathcal{K}^t_{od}} x^t_{oijd} \le 1 \qquad \forall (o,d),\ t \tag{32}$$

$$\sum_j x^t_{oijd} \le y_i, \qquad \sum_i x^t_{oijd} \le y_j \qquad \forall (o,d),\ i \text{ or } j,\ t \tag{33}$$

$$\text{capacity per period (optional)} \qquad \forall i,\ t \tag{34}$$

$$y_i \in \{0,1\}, \quad 0 \le x^t_{oijd} \le 1 \tag{35}$$

G is the number of operating days per year, so both terms are yearly. The network y is shared across periods; only the routing x^t changes. Figure 6 illustrates the structure.

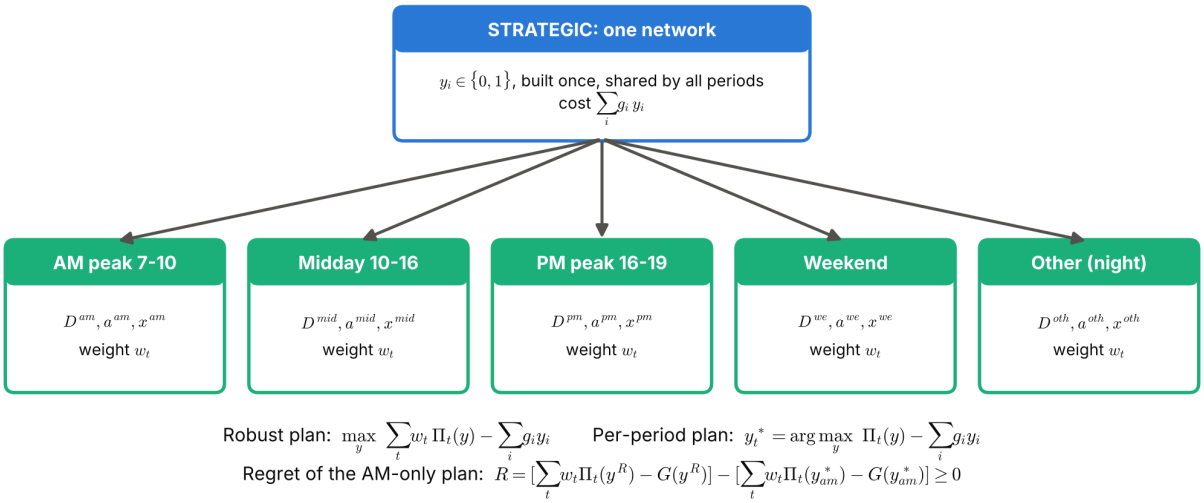


Figure 6: Structure of Model P2 and the comparison with per-period plans.

8 Model properties

Proposition 1 (The routing is integral for a fixed network)**.** *For any fixed $y \in \{0,1\}^{\mathcal{V}}$, Model P1 (without the capacity constraint) has an optimal solution in which each OD pair uses at most one vertiport pair, with $x_{oijd} \in \{0,1\}$. The optimal routing sends OD pair (o,d) through*

$$(i^*, j^*) \in \arg\max\{\pi_{oijd}\, a_{oijd} : (i,j) \in \mathcal{K}_{od},\ y_i = y_j = 1\}$$

if that maximum is positive, and keeps it on the ground otherwise.

Sketch. With y fixed, the problem separates by OD pair. For one pair, the linking constraints force $x_{oijd} = 0$ whenever $y_i = 0$ or $y_j = 0$ and are otherwise implied by the assignment constraint. The remaining feasible set $\{x \ge 0 : \sum x \le 1\}$ is a simplex, so a linear objective is maximized at a vertex: either all zeros or a single option at 1. □

Remark. Proposition 1 gives a closed form for the value of a given network,

$$\Pi(y) = \sum_{(o,d)} D_{od} \max\Big\{0, \max_{(i,j)\in\mathcal{K}_{od}:\, y_i=y_j=1} \pi_{oijd}\, a_{oijd}\Big\} - \sum_i g_i y_i,$$

which makes it cheap to evaluate any network under any demand scenario (needed for the RQ1 and RQ2 metrics below).

Proposition 2 (Strength of the linking constraints)**.**
The OD-aggregated linking constraint $\sum_j x_{oijd} \le y_i$ implies the disaggregated constraints $x_{oijd} \le y_i$ for every j, but not conversely. The formulation is therefore at least as tight as the classical "strong" facility-location formulation.

Structure and size. P1 is a facility location problem in which each customer (OD pair) needs *two* open facilities, which is closer to hub location than to the uncapacitated facility location problem; both are NP-hard in general. For Chicago, $|\mathcal{V}| \approx 20$–30 binaries and $\sum_t \sum_{od} |\mathcal{K}^t_{od}|$ on the order of 10^4–10^5 continuous variables; Gurobi solves instances of this size in seconds to minutes.

9 Evaluation metrics

Let $\Pi^S(y)$ denote the value of network y evaluated under demand scenario S (with optimal routing, Proposition 1).

Definition 1 (Value of heterogeneity, RQ1)**.** Let y^{het} be optimal when shares use (μ_{od}, σ_{od}) and y^{unif} optimal when shares use $(\bar{\mu}, \bar{\sigma})$. With the heterogeneous model taken as the truth,

$$\text{VOH} = \Pi^{\text{het}}(y^{\text{het}}) - \Pi^{\text{het}}(y^{\text{unif}}) \ge 0, \qquad \text{VOH\%} = \frac{\text{VOH}}{\Pi^{\text{het}}(y^{\text{het}})}.$$

It measures how much an operator loses by planning with one average expectation. It is reported for every point of the grid $\mathcal{G}$.

Definition 2 (Regret of a single-period plan, RQ2)**.** Let y^R solve Model P2 and y^*_t solve P1 with the demand of period t only. The regret of planning for the morning peak is

$$\text{Reg(am)} = \Big[\sum_t w_t\, \Pi_t(y^R) - G\sum_i g_i y^R_i\Big] - \Big[\sum_t w_t\, \Pi_t(y^*_{\text{am}}) - G\sum_i g_i y^*_{\text{am},i}\Big] \ge 0.$$

Network similarity. For two networks y, y': the Jaccard index $|S \cap S'|/|S \cup S'|$ of their site sets, and for each site of y' the distance to the nearest site of y (used for the comparison with the paper and with Volakakis and Mahmassani 2024).

Benchmark: p-median. Choose p sites minimizing demand-weighted access distance,

$$\min \sum_o \sum_i D_o\, \ell_{oi}\, z_{oi} \quad \text{s.t.} \ \sum_i z_{oi} = 1,\ z_{oi} \le y_i,\ \sum_i y_i = p,\ y \in \{0,1\},\ z \ge 0,$$

then evaluate its network with $\Pi(y)$ (the paper reports 22–53% lower profit for this benchmark).

10 Relation to the original model

Component	Wang, Jacquillat and Vaze (2022)	This project
Vertiport decisions	location and capacity level z_{ih}, fleet Γ	location y_i (capacity optional)
Matching	y_{oijd} binary, π_{oijd} fractions	x^t_{oijd} (integral at optimum)
Pooling wait	endogenous, $\delta_{ij} \geq (n-1)/\chi_{ij}$ (second-order cone)	fixed $\bar{\delta}$
Detours, reliability	endogenous (queueing network, M/M/1)	not modeled
Fleet, repositioning, charging	modeled	not modeled
Demand function	DRO or MNL S-curve inside the model	exact share (main) and DRO S-curve (comparison), pre-computed; optional transfer penalty h
Customer expectations	uniform μ (0.40), σ	**uniform and OD-specific from fares (RQ1)**
Time of day	morning commute only	**five periods, robust network (RQ2)**
Solution method	adaptive discretization (MIS-OCP)	MILP, Gurobi

The simplified model corresponds roughly to the paper's "exogenous operations" benchmark. The natural extension is to restore the pooling term: with passenger flow $\chi_{ij} = \sum_{(o,d)} D_{od}\, a_{oijd}\, x_{oijd}$, impose $\delta_{ij} \geq (n-1)/(\chi_{ij} + \varepsilon)$ as a second-order cone constraint (Lemma 1 of the paper). The share a then depends on decisions, and a static piecewise-constant discretization of the share (the paper's Sta(M_C) benchmark) keeps the model solvable.

11 Changes in version 1.1

Version 1 was written before the Chicago runs. The following elements were added or changed after the first results (late October to early November 2026).

11.1 Exact switching share (main) and DRO bound (comparison)

Write the per-minute premium of flying on a trip as $X = (P_{od} - q)/t$, so that $\theta = X/b$. A traveler flies on an option with relative saving s if and only if $X \leq s\, b$. With X normal with mean $\mathbb{E}[X]$ and standard deviation $\mathrm{sd}(X)$ on each OD pair (from the five sufficient statistics, Section 5) and b lognormal with median b_0 and coefficient of variation 0.84, independent of X, the share is

$$a^{\text{exact}}_{oijd} = \mathbb{E}_b\left[\Phi\left(\frac{s_{oijd}\, b - \mathbb{E}[X]}{\mathrm{sd}(X)}\right)\right], \tag{36}$$

computed with 24-point Gauss–Hermite quadrature over $\ln b$. For thin OD pairs, $\mathrm{sd}(X)$ is shrunk toward the period median with a prior weight of 30 trips. The paper's distributionally robust (Cantelli) share of Section 4.4 is kept as a comparison. With data-based expectations ($\bar{\mu} > 1$ in most scenarios) it is zero on almost every option, because it is zero whenever $s < \mu$.

Pooled benchmark (RQ1). The average-customer model gives every OD pair the trip-weighted mixture of all OD distributions: $a^{\text{uni}}(s) = \sum_{od} w_{od}\, a_{od}^{\text{exact}}(s) / \sum_{od} w_{od}$. It contains the same travelers as the OD-specific model; only the route identity is removed.

11.2 Minimum time saving and transfer penalty

An option is offered only if $s_{oijd} \geq s_{\min}$ with $s_{\min} = 0.15$ in the main runs. This removes options where switching is driven by price alone (surged ground fares above the air-taxi fare with almost no time saved). The utility of Section 4.4, like the paper's equations (19)–(20), has no fixed cost for changing vehicles. Because the paper's expectations come from surveys, they absorb such costs implicitly; the data-based expectations used here do not. The transfer penalty h enters τ above and is varied over $\{0, 5, 10, 15\}$ minutes, with $s_{\min} \in \{0.15, 0\}$.

11.3 Vertiport capacity

Constraint (29) is applied in Model P2 for every period t with $H_t = 3$, 6, 3, 12 and 24 hours (AM, midday, PM, evening, weekend), demand spread evenly within each period, and $Q \in \{\infty, 200, 100, 50\}$ passengers per hour (departing plus arriving). Single-period networks are evaluated with y fixed, since capacity breaks the closed form of Proposition 1.

11.4 Exact evaluation by enumeration and day-level bootstrap

Without capacity, Proposition 1 gives the value of every network in closed form. With 16 candidates and $p \leq 8$ there are 39,186 networks, so all of them are evaluated directly; this reproduces the Gurobi solutions of P1 and P2 exactly. The 22 weekdays and 9 weekend days of March 2026 are resampled with replacement 200 times (separately by day type). For each resample, demand, ground times, options and shares are rebuilt and all networks are re-evaluated. Reported intervals are the 5th and 95th percentiles across resamples. They measure sampling variation within March-like conditions only, not uncertainty in the assumed parameters.

11.5 Seasonal check

The same pipeline is applied to July 2025 and January 2026 to test whether the network planned on March data remains good in other seasons (cross-season regret and an all-season network with months weighted equally).

12 Scenario plan

Run	Content	Model
V1	NYC validation, μ sweep 0.40 to 0.33, calibrated c (done)	P0
C1	Chicago base case, AM peak, uniform μ, σ	P1
C2	RQ1: OD-specific (μ_{od}, σ_{od}) over the grid $r \in \{2, 3, 4.5, 6, 9\}$, $b \in \{20, 30, 45, 65\}$; VOH	P1
C3	RQ2: per-period networks, robust network, regret	P1, P2
C4	p-median benchmark	P1 evaluation
S	Sensitivity: v, t_{oh}, $\bar{\delta}$, σ, k, $t_{\max}$, price definition (fare only)	all
X	Optional: March 2019 comparison; pooling extension	P1, P2

A NYC validation results (Model P0)

Yellow taxi, March 2019, weekdays 7–10 am, demand scaled by $S = 2$; 26 candidate zones; $v = 240$ km/h, $t_{\text{oh}} = 10$, $\bar{\delta} = 5$, $\sigma = 0.20$; $c = 36.75$ passengers (3 vertiports at $\mu = 0.40$ for any $c \in [32, 41.5]$).

μ	ours	paper	passengers	change in our network
0.40	3	3	125.6	JFK, Times Sq/Theatre District, Gramercy
0.39	3	3	136.6	–
0.38	4	3	187.0	+ Upper East Side North
0.37	4	4	201.9	–
0.36	6	4	290.3	+ Kips Bay, LaGuardia, TriBeCa; − Gramercy
0.35	6	7	316.9	–
0.34	7	9	382.8	+ Gramercy, Midtown Center; − Kips Bay
0.33	7	10	416.7	–

The number of vertiports tracks the paper. Most site differences are adjacent zones (Times Sq vs Clinton East). The model places the LaGuardia-area vertiport at the airport, while the paper places it in Astoria for operational reasons (shorter flights, higher reliability, shorter detours; Section 7.1.1 of the paper): the gap isolates the effect of the operations layer that this model omits.